INCH-POUND

MIL-W-23351/4C
20 January 2009
SUPERSEDING
MIL-W-23351/4B
20 October 1977

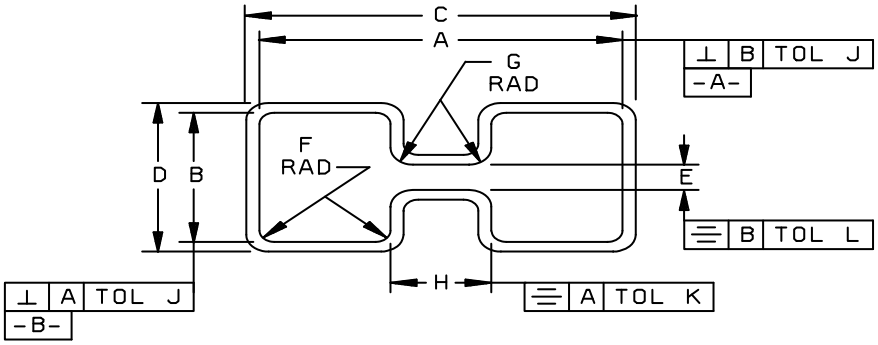
DETAIL SPECIFICATION SHEET

WAVEGUIDES, DOUBLE RIDGE
(BANDWIDTH RATION 2.4:1)

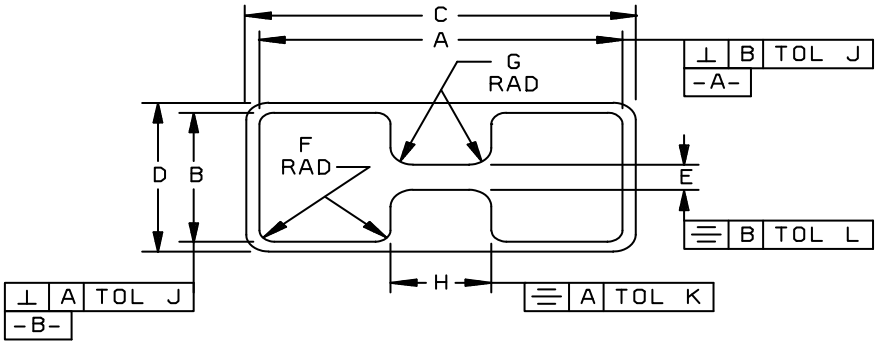
This specification is approved for use by all Departments and Agencies of the Department of Defense.

Inactive for new design
after 8 May 1998

The complete requirements for procuring the waveguides described herein shall consist of this specification sheet and MIL-DTL-23351.



CONFIGURATION 1



CONFIGURATION 2

FIGURE 1. Waveguide.

AMSC N/A

FSC 5985

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TABLE I. Part number dimensions and material.

Part number M23351/4-	Dimensions 1/				
	A	B	C	D	E
001 2/ 002 003 004	29.667± .020 (753.54) (.51)	13.795 ± .020 (350.39) (.51)	29.917 Max (759.89)	14.045 Max (356.74)	5.863 ± .010 (148.92) (.25)
005 006 007 008	19.428 ± .015 (493.47) (.38)	9.034 ± .015 (229.46) (.38)	19.678 Max (499.82)	9.284 Max (235.81)	3.839 ± .007 (97.51) (.18)
009 010 011 012	12.33 ± 0.10 (313.26) (.25)	5.737 ± .010 (145.72) (.25)	12.583 ± .015 (319.61) (.38)	5.987 ± .015 (152.07)(.38)	2.437 ± .005 (61.90) (.13)
013 014 015 016 2/	8.100 ± .010 (205.74) (.25)	3.767 ± .010 (95.68) (.25)	8.350 ± .015 (212.09) (.38)	4.017 ± .015 (102.03) (.38)	1.601 ± .004 (40.67) (.10)
017 3/ 018 019 020	6.167 ± .005 (156.64) (.13)	2.868 ±.005 (72.85) (.13)	6.417 ± .005 (162.99) (.13)	3.118 ± .005 (79.20) (.13)	1.219 ± .003 (30.96) (.08)
021 022 023 024	3.455 ± .004 (87.76) (.10)	1.607 ± .004 (40.82) (.10)	3.615 ± .004 (91.82) (.10)	1.767 ± .004 (44.88) (.10)	.6830 ± .003 (17.348) (.08)
025 026 027 028	2.590 ± .004 (69.79) (.10)	1.205 ± .004 (30.61) (.10)	2.750 ± .004 (69.85) (.10)	1.365 ± .004 (34.67) (.10)	.512 ± .002 (13.00) (.05)
029 030 031 032	1.480 ± .003 (37.59) (.08)	.688 ± .003 (17.48) (.08)	1.608 ± .004 (40.84) (.10)	.816 ± .004 (20.73) (.10)	.2920 ± .002 (7.417) (.05)
033 034 035 036	1.090 ± .003 (27.69) (.08)	.506 ± .003 (12.48) (.08)	1.190 ± .003 (30.23) (.08)	.606 ± .003 (15.39)(.08)	.215 ± .002 (5.46) (.05)
037 038 039 040	.691 ± .003 (17.55) (.08)	.321 ± .003 (8.15) (.08)	.791 ± .003 (20.09) (.08)	.421 ± .003 (10.69) (.08)	.136 ± .002 (3.45) (.05)
041 042 043 044	.471 ± .003 (11.96) (.08)	.219 ± .003 (5.56) (.08)	.551 ± .003 (14.00)(.08)	.299 ± .003 (7.59) (.08)	.0930 ± .002 (2.362) (.05)
045 046 047 048 3/	.288 ± .003 (7.32) (.08)	.134 ± .003 (3.40) (.08)	.368 ± .003 (9.35) (.08)	.214 ± .003 (5.44) (.08)	.0570 ± .002 (1.448) (.05)
049 3/ 4/ 050 3/ 4/ 051 3/ 4/ 052 3/ 4/	1.025 ± .003 (26.04) (.08)	.475 ± .003 (12.06) (.08)	1.125 ±.005 (28.58) (.13)	.575 ± .005 (14.60) (.13)	.191 ± .003 (4.85) (.08)

See footnotes at end of table I

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TABLE I. Part number dimensions and material – Continued.

Part number M23351/4-	Dimensions (continued) 1/						
	F Max	G ± 10%	H	J	K	L	Material
001 2/	.063 (1.60)	1.173 (29.79)	7.417 ± .010 (188.39) (.25)	.020 (.51)	.010 (.25)	.010 (.25)	Aluminum alloy
002							Copper alloy
003							Copper
004							Silver alloy
005	.050 (1.27)	.768 (19.50)	4.857 ± .007 (123.37) (.18)	.015 (.38)	.007 (.18)	.007 (.18)	Aluminum ally
006							Copper alloy
007							Copper
008							Silver alloy
009	.050 (1.27)	.487 (12.37)	3.083 ± .005 (78.31) (.13)	.010 (.25)	.005 (.13)	.005 (.13)	Aluminum alloy
010							Copper alloy
011							Copper
012							Silver alloy
013	.050 (1.27)	.320 (8.13)	2.025 ± .004 (51.44) (.10)	.010 (.25)	.004 (.10)	.004 (.10)	Aluminum alloy
014							Copper alloy
015							Copper
016 2/							Silver alloy
017 3/	.050 (1.27)	.244 (6.20)	1.542 ± .003 (39.17) (.08)	.005 (.13)	.003 (.08)	.003 (.08)	Aluminum alloy
018							Copper alloy
019							Copper
020							Silver alloy
021	.050 (1.27)	.137 (3.48)	.8640 ± .003 (21.946) (.08)	.004 (.10)	.003 (.08)	.003 (.08)	Aluminum alloy
022							Copper alloy
023							Copper
024							Silver alloy
025	.050 (1.27)	.102 (2.59)	.648 ± .002 (16.46) (.05)	.004 (.10)	.002 (.05)	.002 (.05)	Aluminum alloy
026							Copper alloy
027							Copper
028							Silver alloy
029	.030 (.76)	.058 (1.47)	.3700 ± .002 (9.398) (.05)	.003 (.08)	.002 (.05)	.002 (.05)	Aluminum alloy
030							Copper alloy
031							Copper
032							Silver alloy
033	.030 (.76)	.043 (1.09)	.272 ± .002 (6.91) (.05)	.003 (.08)	.002 (.05)	.002 (.05)	Aluminum alloy
034							Copper alloy
035							Copper
036							Silver alloy
037	.020 (.51)	.027 (.69)	.173 ± .002 (4.39) (.05)	.003 (.08)	.002 (.05)	.002 (.05)	Aluminum alloy
038							Copper alloy
039							Copper
040							Silver alloy
041	.015 (.38)	.019 (.48)	.1180 ± .002 (2.997) (.05)	.003 (.08)	.002 (.05)	.002 (.05)	Aluminum alloy
042							Copper alloy
043							Copper
044							Silver alloy
045	.015 (.38)	.011 (.28)	.0720 ± .002 (1.829) (.05)	.003 (.08)	.002 (.05)	.002 (.05)	Aluminum alloy
046							Copper alloy
047							Copper
048 3/							Silver alloy
049 3/ 4/	.030 (.76)	.062 (1.57)	.256 ± .003 (6.50) (.08)	.003 (.08)	.003 (.08)	.003 (.08)	Aluminum alloy
050 3/ 4/							Copper alloy
051 3/ 4/							Copper
052 3/ 4/							Silver alloy

1/ Dimensions are in inches. Meter equivalents (in parentheses) are given for general information only and are based upon 1.00 inch = 25.4 mm.
2/ The waveguide shall be in accordance with configuration 1.
3/ The waveguide shall be in accordance with configuration 2.
4/ Eccentricity shall be .004 maximum.

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REQUIREMENTS:

Design, Construction, and physical dimension: See figure 1 and table I
Material: See table I.
Part number: M23351/4- (dash number from table I).

NOTES:

Engineering information and frequency range: See table II.
Cross reference: See table III.

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TABLE II. Engineering information.

Part number M23351/4-	Suggested frequency range TE_{10} mode	Cutoff frequency		Theoretical attenuation at f = √3 f_{c10} 1/ 2/	Theoretical power handling capacity at f = √3 f_{c10} 3/
		TE_{10} mode	TE_{20} mode		
	GHz	GHz	GHz	db/ft	Kw
001-xxx	0.175-0.420	0.146	0.430	0.00035	61,960
002-xxx				0.00034	
003-xxx				0.00023	
004-xxx				0.00025	
005-xxx	0.267-0.640	0.222	0.657	0.00065	26,570
006-xxx				0.00064	
007-xxx				0.00043	
008-xxx				0.00046	
009-xxx	0.420-1.000	0.350	1.034	0.00128	10,710
010-xxx				0.00126	
011-xxx				0.00085	
012-xxx				0.00091	
013-xxx	0.640-1.530	0.533	1.575	0.0024	4,620
014-xxx				0.0024	
015-xxx				0.0016	
016-xxx				0.0017	
017-xxx	0.840-2.000	0.700	2.068	0.0036	2,676
018-xxx				0.0036	
019-xxx				0.0024	
020-xxx				0.0026	
021-xxx	1.500-3.600	1.249	3.692	0.0087	840.5
022-xxx				0.0086	
023-xxx				0.0058	
024-xxx				0.0062	
025-xxx	2.000-4.800	1.666	4.925	0.0134	472.5
026-xxx				0.0132	
027-xxx				0.0089	
028-xxx				0.0095	
029-xxx	3.500-8.200	2.915	8.620	0.0307	151.3
030-xxx				0.0303	
031-xxx				0.0204	
032-xxx				0.0218	
033-xxx	4.750-11.000	3.961	11.705	0.0487	83.72
034-xxx				0.0481	
035-xxx				0.0324	
036-xxx				0.0347	
037-xxx	7.500-18.000	6.239	18.464	0.0964	33.58
038-xxx				0.0951	
039-xxx				0.0641	
040-xxx				0.0686	
041-xxx	11.000-26.500	9.363	27.080	0.171	15.63
042-xxx				0.169	
043-xxx				0.114	
044-xxx				0.122	
045-xxx	18.000-40.000	14.995	44.285	0.358	5.834
046-xxx				0.353	
047-xxx				0.238	
048-xxx				0.255	
049-xxx	4.75-11.00	4.086	12.07	-----	-----
050-xxx					
051-xxx					
052-xxx					

See footnotes at end of table II

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1/ $f_{c_{10}}$ = cutoff frequency for TE_{10} mode.

2/ Actual attenuation of waveguides may be considerable greater depending upon operating frequency and temperature. Typical resistivities of materials (at 20° C): aluminum alloy (6061) = 4.0 microhms-cm; copper alloy (brass) = 3.9 microhms-cm; copper -1.77 microhm –cm; silver alloy (coin silver) = 2.03 microhm-cm.

3/ Actual power handling capability of waveguides may be considerable less depending upon waveguide temperature, altitude, operating frequency, etc. The tabulated values were calculated based upon the E field produced breakdown in a nonpressurized air dielectric waveguide under continuous wave (CW) conditions. The breakdown strength of air was considered to be 15000 volts per centimeter, corner radii considered.

TABLE III. Cross reference (type designation to part number).

Type designation	Part number M23351/4-	Type designation	Part number M23351/4-
WRD175U24A	001	WRD200D24A	025
WRD175U24B	002	WRD200D24B	026
WRD175U24C	003	WRD200D24C	027
WRD175U24S	004	WRD200D24S	028
WRD267U24A	005	WRD350D24A	029
WRD267U24B	006	WRD350D24B	030
WRD267U24C	007	WRD350D24C	031
WRD267U24S	008	WRD350D24S	032
WRD420U24A	009	WRD475D24A	033
WRD420U24B	010	WRD475D24B	034
WRD420U24C	011	WRD475D24C	035
WRD420U24S	012	WRD475D24S	036
WRD640U24A	013	WRD750D24A	037
WRD640U24B	014	WRD750D24B	038
WRD640U24C	015	WRD750D24C	039
WRD640U24S	016	WRD750D24S	040
WRD840U24A	017	WRD110C24A	041
WRD840U24B	018	WRD110C24B	042
WRD840U24C	019	WRD110C24C	043
WRD840U24S	020	WRD110C24S	044
WRD150D24A	021	WRD180C24A	045
WRD150D24B	022	WRD180C24B	046
WRD150D24C	023	WRD180C24C	047
WRD150D24S	024	WRD180C24S	048

Referenced documents. This document only references MIL-DTL-23351.

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Navy – EC
Air Force – 85
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Preparing activity:
DLA – CC
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Review activities:
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